

Evaluation Integrity for AI Systems

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Research Vision

A benchmark score is valuable only when success on the benchmark reflects the underlying capability it was designed to test. For modern language models, that connection is increasingly difficult to guarantee. Models are trained on internet-scale corpora, while benchmarks are published online and can enter future training data. A model may therefore score well because it has encountered the evaluation before rather than because it can solve the underlying task. At the same time, once an evaluation becomes a known target, model developers have an incentive to optimize performance on the evaluation itself. Together, these forces weaken the connection between benchmark performance and evidence of underlying capability. This problem now extends beyond research practice as regulators, auditors, and organizations increasingly use benchmark results to assess and compare AI systems.

I call this problem **evaluation integrity**: whether success on an evaluation can be attributed to the capability under test rather than to advantages created by prior exposure or benchmark-specific optimization. My research addresses this problem under two different conditions. When a benchmark is still private, I study how to estimate a model's performance on it without exposing the benchmark to the model provider, so the benchmark can remain unexposed throughout the evaluation process. When a benchmark is already exposed, I study how to determine whether exposure has distorted the measured performance and how to recover the benchmark's ability to distinguish generalizable capability from recall. Across both conditions lies a longer-term challenge: preserving evaluation integrity over repeated use as models, training data, and sources of exposure continue to change.

Research Foundations

Grounding Evaluation in Observed Behavior. My research on evaluation grew out of studying state reconciliation in infrastructure-as-code (IaC): whether automation moves a computing environment from its current state to the intended one. I conducted an empirical study of 5,110 state-reconciliation defects, using qualitative analysis to develop a defect taxonomy. I hypothesized that the defect patterns observed in practice could guide the discovery of similar failures in an open-source IaC compiler. From these patterns, I derived heuristics for constructing LLM prompts to uncover defects, identifying **nine previously unknown defects** in the Ansible compiler (**FSE 2024**) [1]. These failures directed my attention to the relationship between the automation, the environment in which it executes, and the state it is intended to produce. The lesson I drew from this work extended beyond IaC: correctness should be established from the state a system actually produces, not from the apparent plausibility of the artifact intended to produce it.

That lesson became directly relevant to AI evaluation as large language models (LLMs) began to be used for code generation and automation, where plausible code offers no guarantee of the intended system behavior. However, existing IT automation benchmarks did not adequately test whether generated automation actually produced the state practitioners intended. This led me to ask: *can an LLM translate a real-world IT automation request into executable automation that reliably brings the environment to the desired state?* I addressed this question by leading the development of **ExITBench**, an execution-based benchmark that evaluates LLMs on real-world IT automation tasks by executing their generated solutions in controlled environments and verifying the resulting system state (**Findings of ACL 2026**) [2]. Even with ten attempts, the strongest model we evaluated succeeded on only 23.9% of tasks. Failure analysis revealed that the main failures were not superficial coding mistakes, they concentrated around state-reconciliation reasoning (42.1%) and specialized knowledge of automation modules (26.2%). These findings showed why evaluating generated automation requires checking its effects on the environment, rather than judging the generated code alone.

Together, these projects established the first foundation of my research on evaluation integrity: claims about model capability should be grounded in the behavior that matters, not merely in outputs that appear correct. But observing the right behavior is only the first step; it does not yet tell us how broadly that behavior generalizes or whether success reflects capability rather than prior exposure.

When Benchmark Scores Can Be Trusted. Execution can verify that a model produced the intended outcome, but even a correct result may provide only a partial view of the capability we think we are measuring. In collaborative work on program reasoning (ACL 2026), we showed that success on one form of reasoning does not reliably generalize to another [3]. The gap persisted with larger models and reasoning-focused post-training, suggesting that scale alone does not make these forms of reasoning interchangeable. This revealed that even a valid benchmark score may support only a narrow capability claim. This still assumes that the score comes from the model’s underlying ability. For modern language models, that assumption can fail when the benchmark has been exposed during training. This led me to ask: *how can we evaluate a black-box model on a private benchmark without revealing the benchmark and creating the very exposure that would undermine the score?*

I addressed this question through developing the **Private Evaluation Protocol (PEP)** (AACL-IJCNLP 2026), which estimates performance on a private benchmark from limited interaction with a black-box model [4]. On structured benchmarks, PEP estimated target-model accuracy within 5.88 percentage points on average while keeping the private remainder hidden from the model provider. Reasoning-heavy tasks were more challenging, suggesting that limited observations may not always capture enough of a model’s behavior to predict its performance on unseen examples. PEP therefore established that private evaluation is possible without surrendering the benchmark, but also revealed the central tension that remains: protecting the benchmark limits how much behavior we can observe, while reliable evaluation depends on observing enough.

Together, these projects shifted my research from asking whether an evaluation observes the right behavior to asking whether its score can still be attributed to the capability under test. The next step is to understand how that attribution can be preserved, recovered, and sustained as evaluations and models evolve.

Future Research Agenda

My prior work shows that reliable evaluation requires more than observing whether a model succeeds: we must also understand what that success establishes and whether it can still be attributed to the capability under test. My future research develops evaluation integrity across the full life-cycle of a benchmark, from preserving private evaluations to recovering exposed ones and maintaining both over time.

Evaluation Without Exposure

PEP showed that behavior observed on accessible examples can predict performance on hidden ones. My future work asks a broader question: *under what conditions does behavior observed on accessible tasks provide reliable evidence about behavior on a protected evaluation?* One challenge is **what behavior to compare**: for generation, code, tool use, and agentic tasks, matching answers is insufficient, so I will develop task-specific measures of behavioral agreement. Another is **which behavior to observe**: rather than reveal examples at random, I will identify the interactions that are most informative about a target model and use them to build higher-fidelity surrogates. A more ambitious question is **how far that evidence can transfer**. I will map benchmarks by the skills and task structures they require and test whether behavior on nearby public benchmarks can predict performance on a protected benchmark or strengthen its surrogate. When such transfer is weak, targeted interaction with the protected benchmark remains necessary.

Diagnosing and Repairing Exposed Benchmarks

For an exposed benchmark, the central question is not merely whether the model saw the evaluation, but whether that exposure changed its behavior and made the score overstate its capability. I want to understand how this effect depends on the evaluation: *the same exposure may leave different behavioral signatures across tasks and modalities, and exposure need not create a measurable advantage*. A major challenge is that signals such as likelihood or perplexity, accuracy, confidence, and output diversity may not carry the same meaning across settings. Another is distinguishing familiarity from task difficulty, calibration, or ordinary generalization. My goal is to identify which evidence indicates an exposure-driven advantage and which signals remain informative across benchmarks.

Repair raises a parallel question: *how should an exposed benchmark be transformed when the effects of exposure differ across evaluation modalities?* A transformation that preserves meaning in text may have no meaningful analogue in code or vision, and a strategy that removes an exposure advantage in one setting may fail in another. I will study what meaning-preserving perturbation should mean across evaluation modalities, whether successful mitigation strategies transfer between them, and where that transfer breaks down. Most importantly, I will test whether these differences are consistent with the modality-specific signatures identified during diagnosis. If the way exposure appears and the way it can be mitigated vary together, that would provide a more unified account of both what benchmark exposure means and how its effects can be reversed.

Maintaining Benchmark Integrity Over Time

Evaluation integrity must survive more than a single model evaluation. My central question is: *how should an evaluation adapt as the models it measures continue to change without giving up the integrity it was designed to preserve?* For a private benchmark, each new model may be harder to approximate with the information already revealed. The challenge is to maintain surrogate fidelity and estimation accuracy across successive models while revealing as little additional benchmark data as possible. Because every newly revealed example is an irreversible loss of privacy, I want to understand when existing evidence is sufficient, when additional revelation becomes necessary, and how to minimize the cumulative information released over the benchmark’s lifetime. A repaired benchmark faces a different maintenance problem. As it is reused and new models are trained, familiarity with the repaired evaluation can accumulate again. The challenge is to recognize when its diagnostic value has begun to erode and determine whether further repair is sufficient or replacement is necessary. I plan to connect these two challenges through **state reconciliation**: characterize the benchmark’s current integrity state, compare it with the state required for reliable evaluation, and intervene only when that gap becomes meaningful. This raises the longer-term question of whether benchmark integrity can be maintained indefinitely under repeated evaluation.

Fit and Funding

This program sits at the intersection of **machine learning** and **software engineering**, with a broader goal of establishing reliable evidence of AI capability. The scientific questions concern how model behavior transfers to protected evaluations, how prior exposure changes what benchmark scores mean, and how evaluation integrity can be preserved as models evolve. Software systems provide a rigorous setting for this agenda because generated solutions can be executed and their effects on system state can be checked. By varying defects and execution conditions, I will investigate whether evaluation scores reflect models’ ability to generate reliable automation, diagnose defects, and repair software, or advantages from familiarity with benchmark instances. My prior collaborations span learning sciences, ECE, and mathematics, giving me a foundation for extending this agenda across disciplinary boundaries. I will build on this experience through an externally funded program targeting **NSF CISE Future CoRe** and **SaTC**, while developing the benchmark-lifecycle direction toward an **NSF CAREER** proposal. Because much of the agenda relies on black-box access, controlled experimentation, and evaluation infrastructure rather than frontier-scale model training, it is well suited to an academic lab and creates research opportunities for students across benchmark construction, execution-based software testing, automated environment design, surrogate modeling, exposure analysis, and benchmark maintenance.

References

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